



Data structure and Programming

TP09: Queue data structure

Deadline:
6 days

- Objective:
 - To practice create Stack data structure that is able to store unlimited data and data is stored via LIFO method
 - Implement push and pop methods for stack data structure.
 - Apply C++ programming
- Individual work
- Submit to Moodle (do not zip)
 - *A pdf report (cover, exercise and solution with screenshot)*
 - *Source codes*

1. Write a C++ program to implement a queue data structure using an array to store integer numbers. Write functions for: enqueue(), dequeue(), front(), isEmpty(), isFull(), displayQueue(), getSize().
2. Write a C++ program to implement a queue data structure using a singly linked list to store integer numbers with the following operations: enqueue(), dequeue(), front(), isEmpty(), isFull(), displayQueue(), getSize().
3. Create a queue that can schedule a set of priority tasks. Each task is just to display a message on the screen. Table 1 shows about tasks and their descriptions (used for displaying message). Suppose that, the queue contains Task4, Task1, Task3, and Task2 (Task 4 is the first task in the queue, Task 1 is the second task in the queue, so on and so forth). Write a program to ask user “Do you want to activate a task in your queue now?”. When the user say ‘yes’, activate the task by calling to dequeue function of the queue. Keep asking user until there is no task remained in the queue.

Table 1: List of tasks and their descriptions

Task	Description
1	Doing task 1 DONE!
2	Doing task 2 DONE!
3	Doing task 3 DONE!
4	Doing task 4 DONE!

4. At a restaurant, customers are waiting to be served. Create a queue as singly linked list that can store customer. Each customer has name, sex and phone number.

The program provides a menu of choices that a user can use to do some operations on the queue. The operations are:

- A. enqueue a customer (add a customer to queue),
- B. dequeue a customer (remove a customer from queue),
- C. display all customers' info in the queue and
- D. exit the program.

Each time the function dequeue is called, display this message "Serving customer, [name]!". The program keeps running until the user choose choice number 4 to exit.

```
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
    Enter a choice: 1
Enter a student's name: Bob
Enter a student's sex: M
Enter a student's phone number: 012123445
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
    Enter a choice: 1
Enter a student's name: Jack
Enter a student's sex: M
Enter a student's phone number: 070897809
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
    Enter a choice: 3

1 Bob    M        012123445
2 Jack   M        070897809
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
    Enter a choice: 3
```

```
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
    Enter a choice: 3

1 Bob    M        012123445
2 Jack   M        070897809
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
    Enter a choice: 2

Serving customer, Bob!
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
    Enter a choice: 3
```

